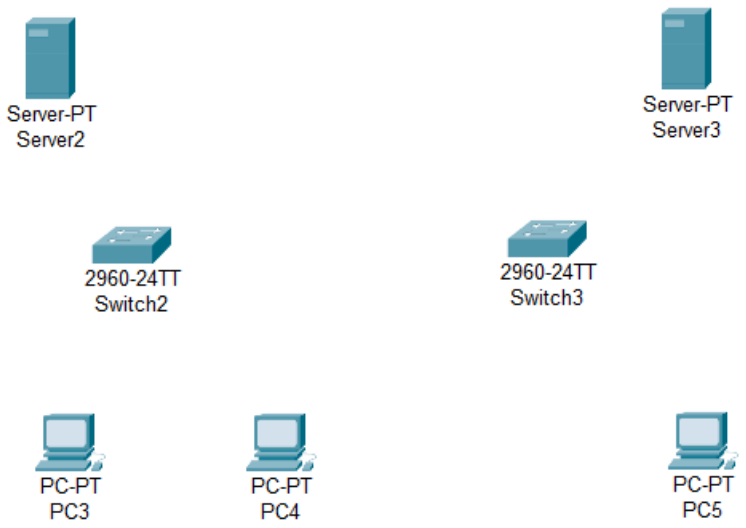
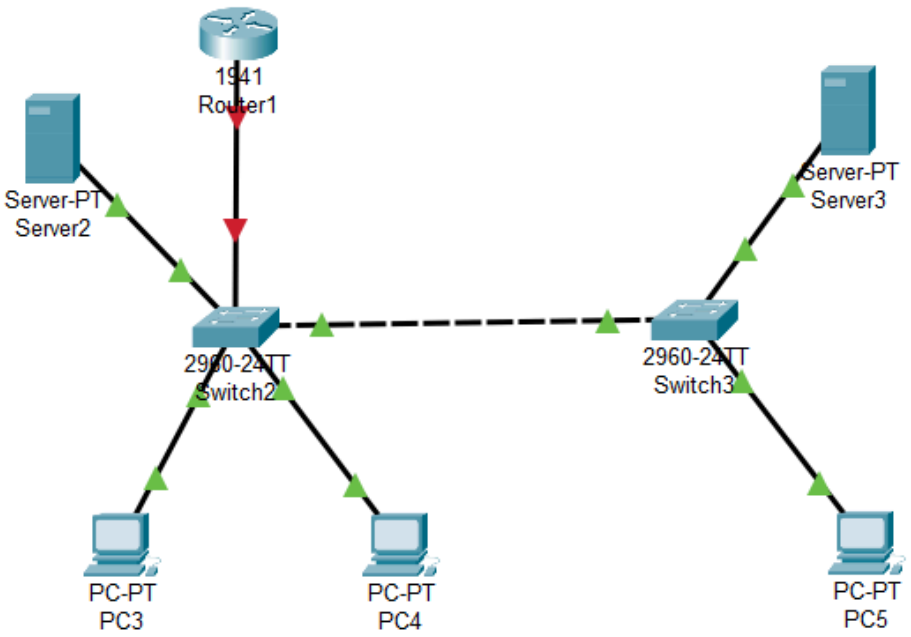
Brandon Gracy 01/09/2025

Today we are going to see a connection between different Vlans.

First of all, let's put our pcs servers and a router.



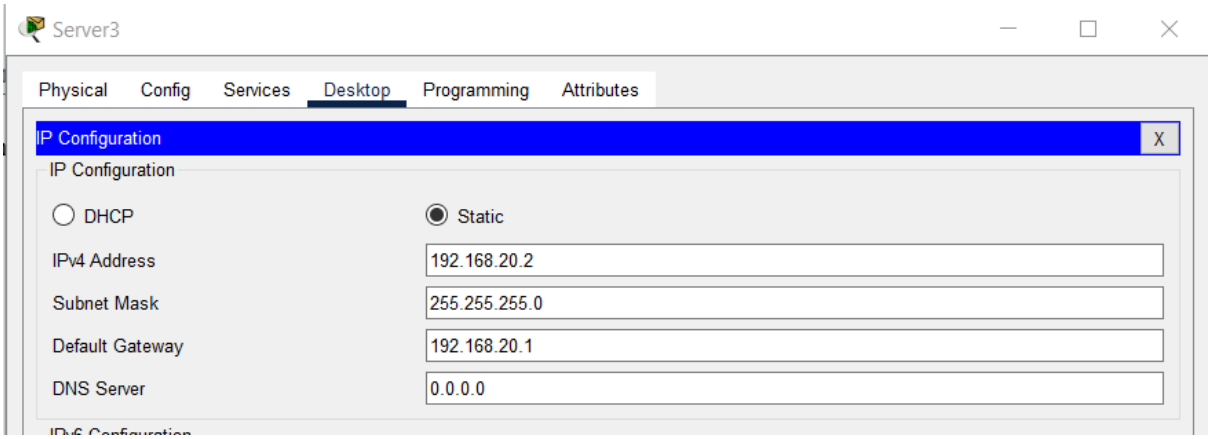
That will be our topology, we have 3 pcs, 2 switches and 1 router, that will be responsible for communication between vlans.

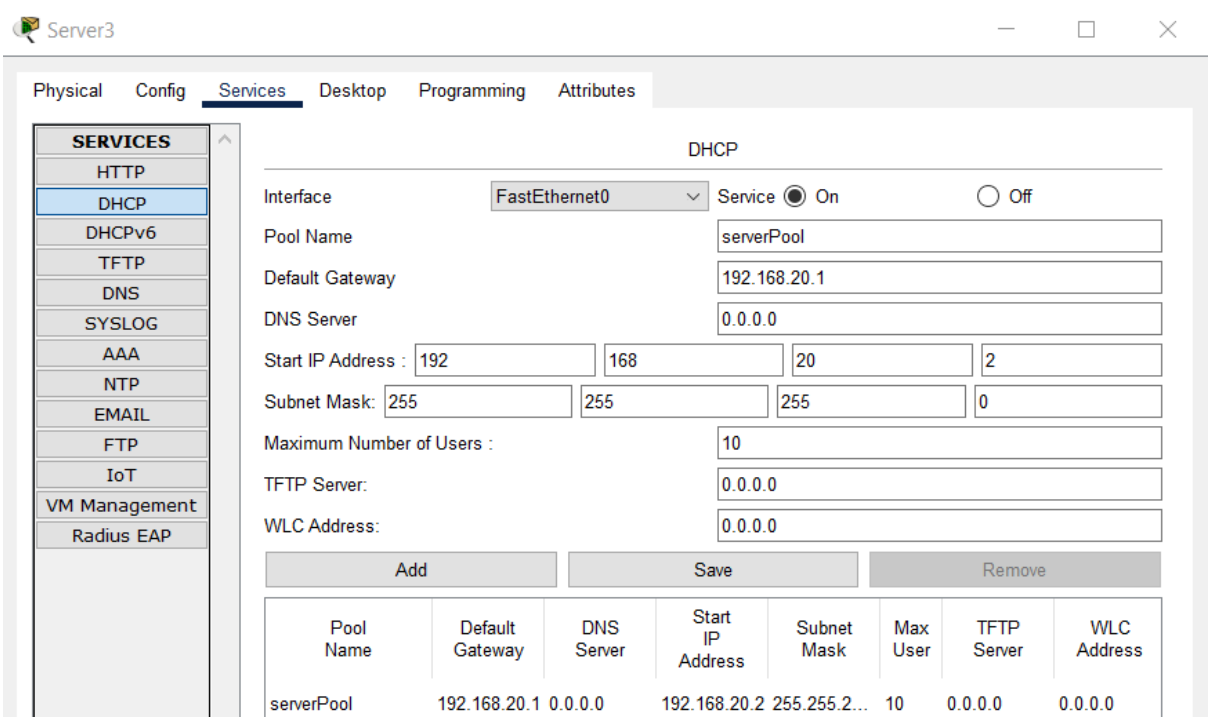


Here I will show the connection of our topology, was used copper straight-through cable to connect pcs in servers, switches and the router, but to connect switch with switch was used copper cross-over.

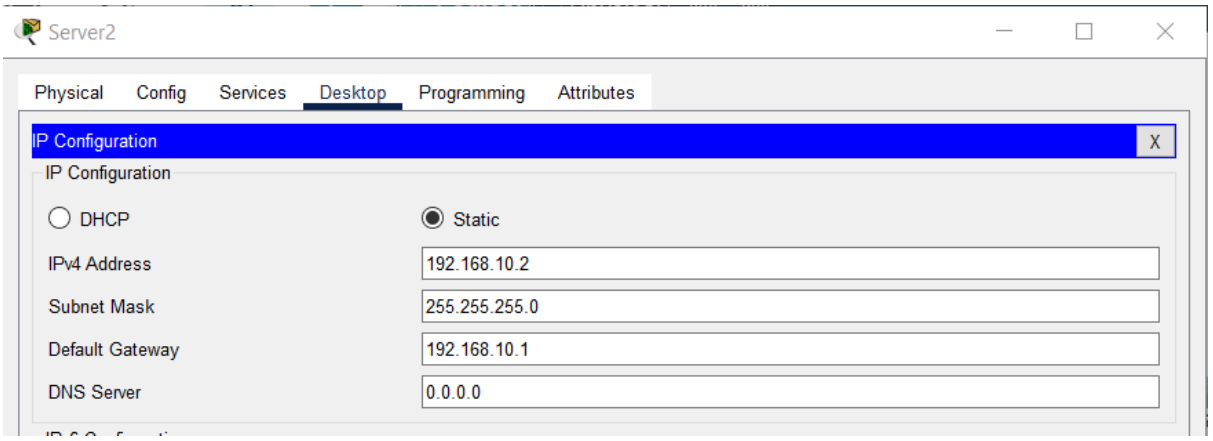
```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

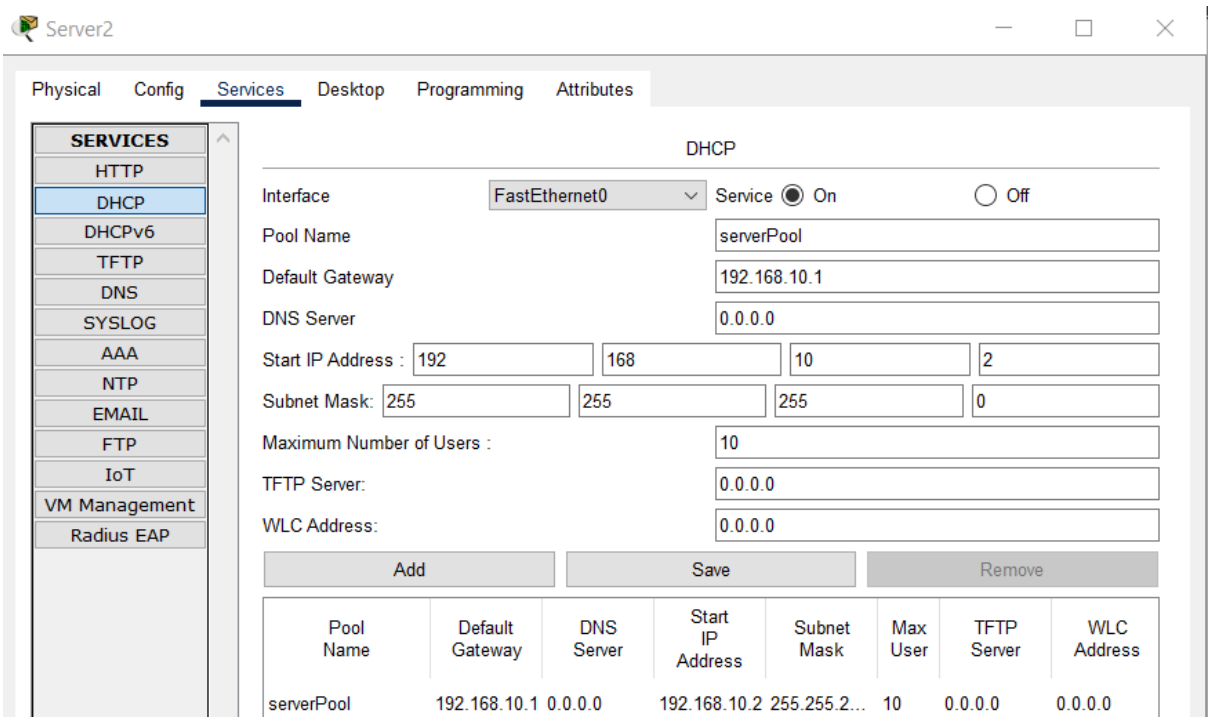
This is a configuration that we need for our switch, done in both switches, switch has to know our vlans to work, and to communicate with each other.



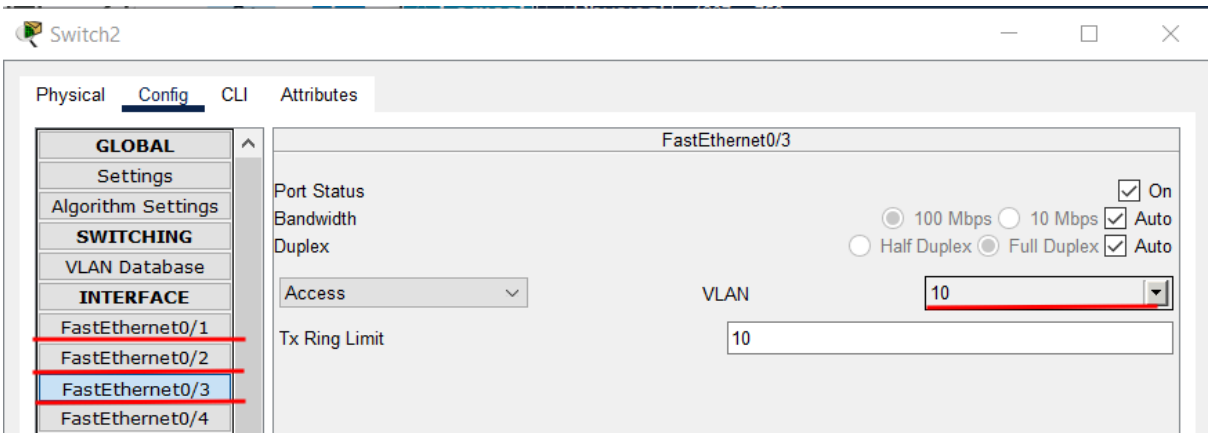


This is the IT server configuration that will be responsive to give IP addresses for our IT vlans.

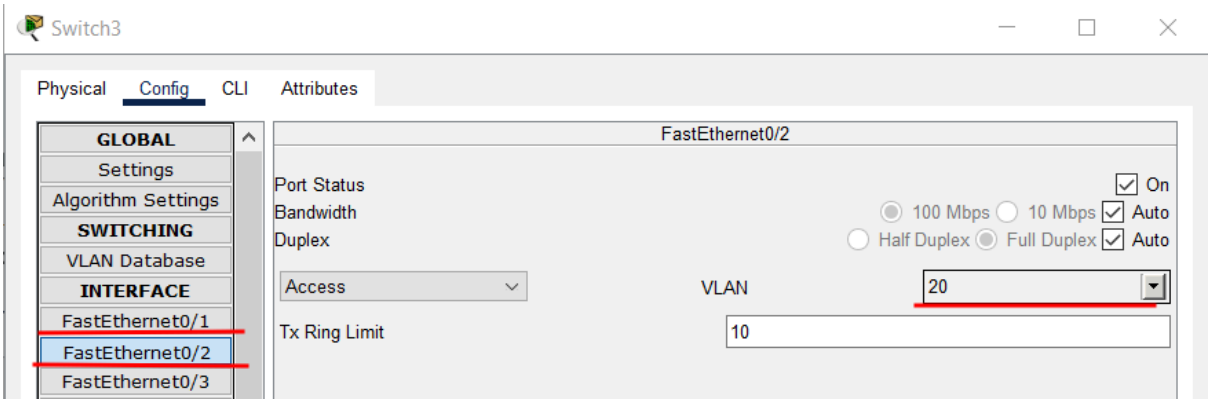




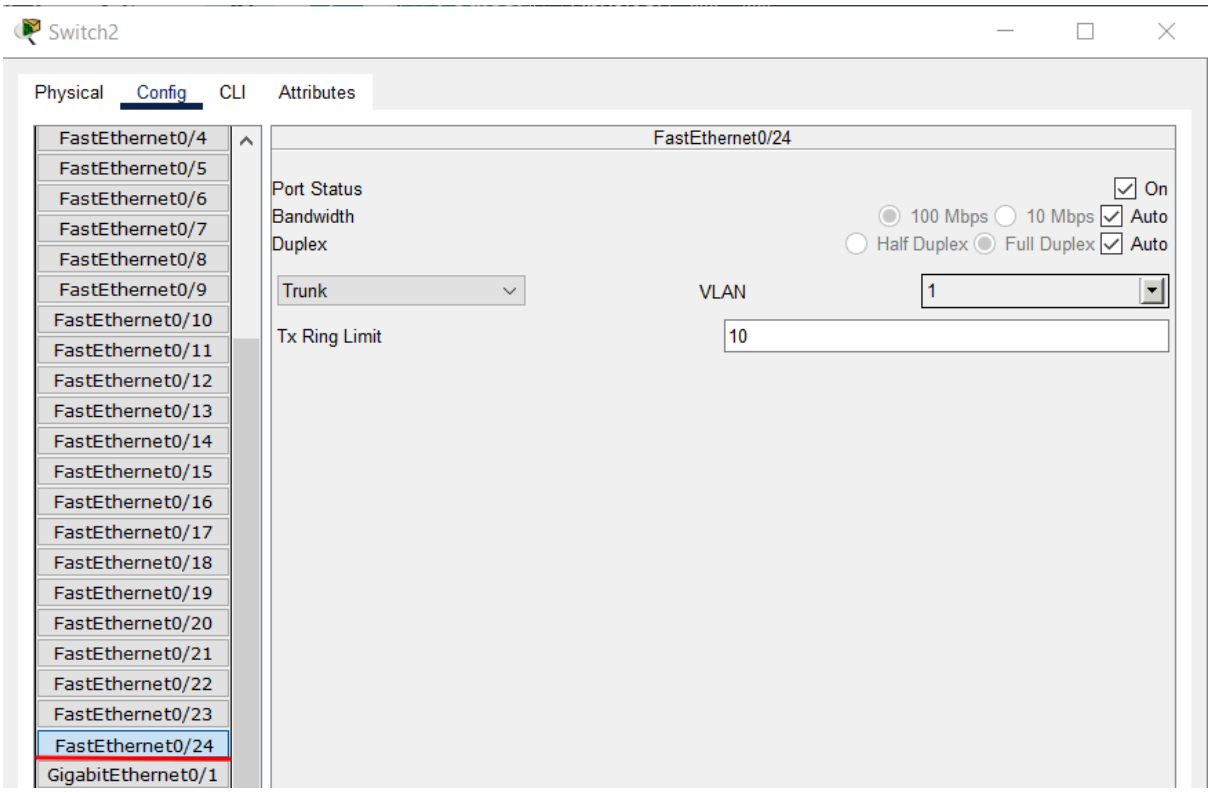
And here, the configuration HR server, that will be responsive to give IP addresses to our HR vlans.



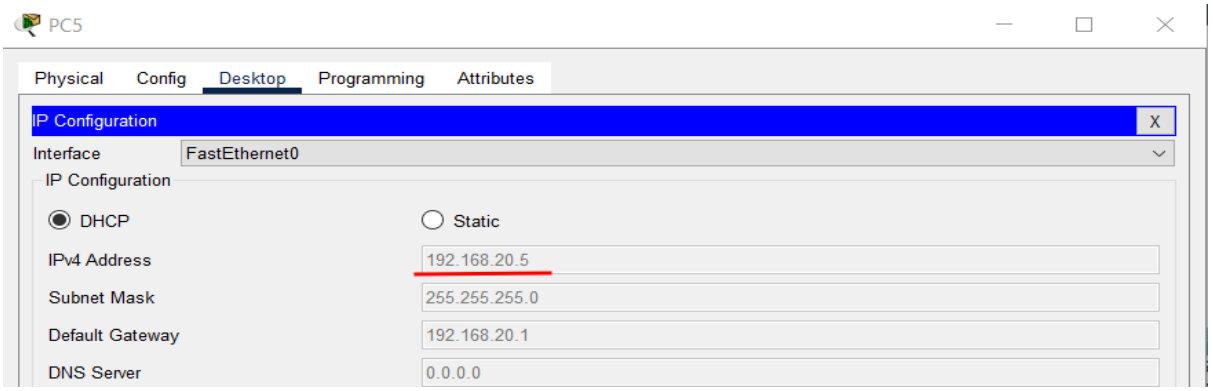
Now we have to put each fastEthernet in your due vlan. how we are using the switch for the HR vlan, lets put in your respective vlan, the vlan 10.

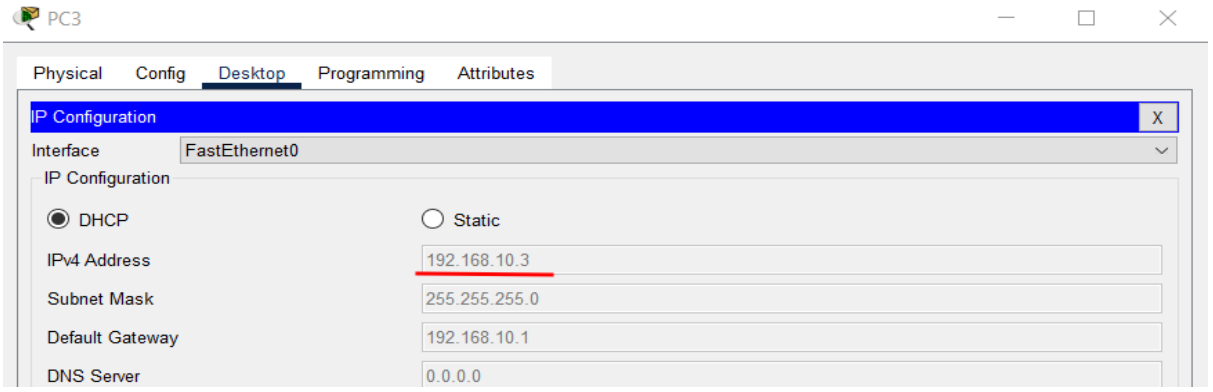


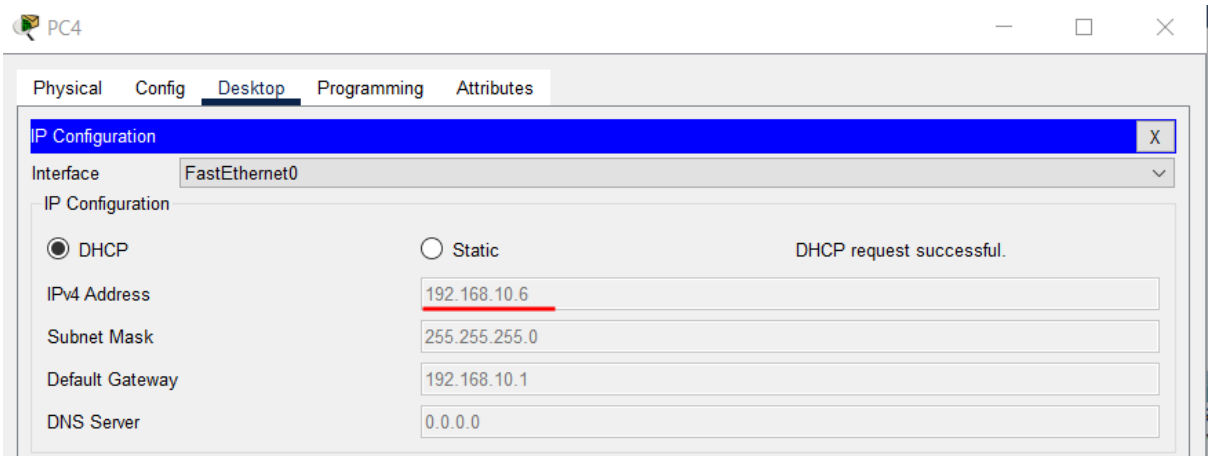
Here we did the same, but in our other switch, the IT vlan.



The output fasterEthernet0/24 has to be in trunk, to successful connection, we are going to do this in both switches.



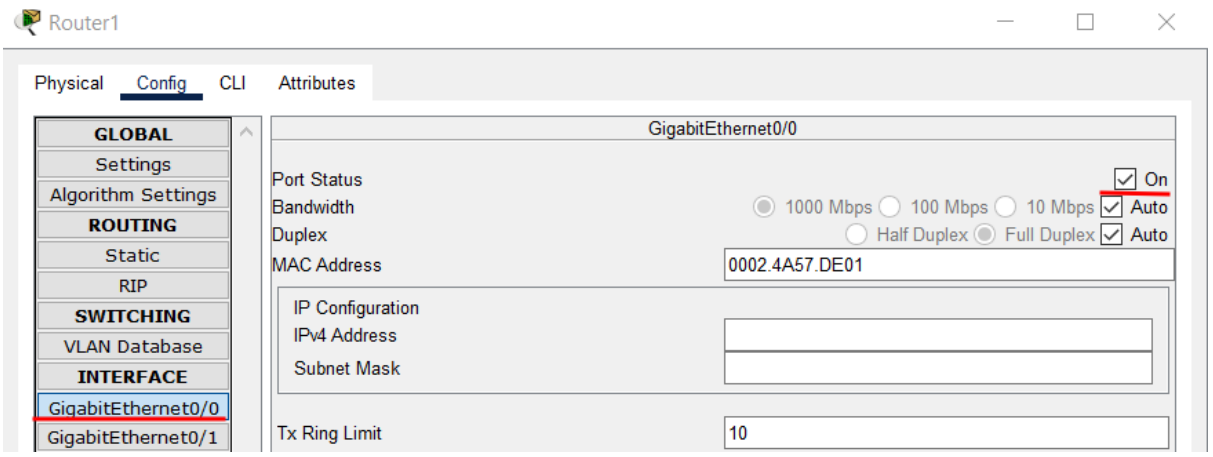




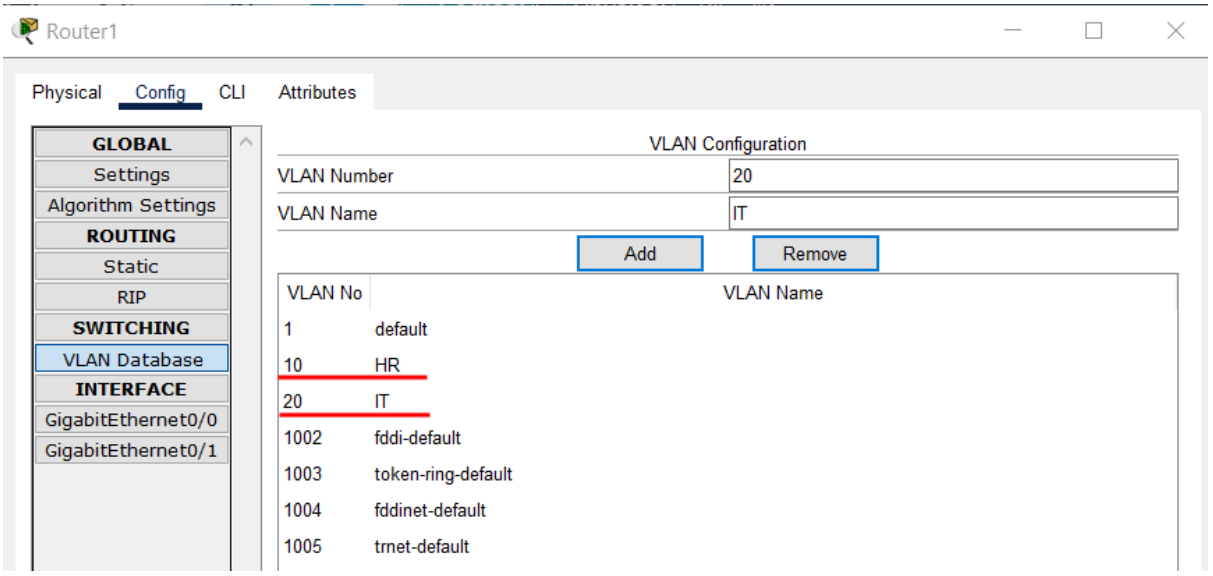
Here we can check that ours DHCP servers its working perfectly. they did their job assign IPs addresses to our pcs.

```
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Gigabitethernet 0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface GigabitEthernet 0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#confi t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet 0/0
Router(config-if)#no shutdown
Router(config-if)#end
```

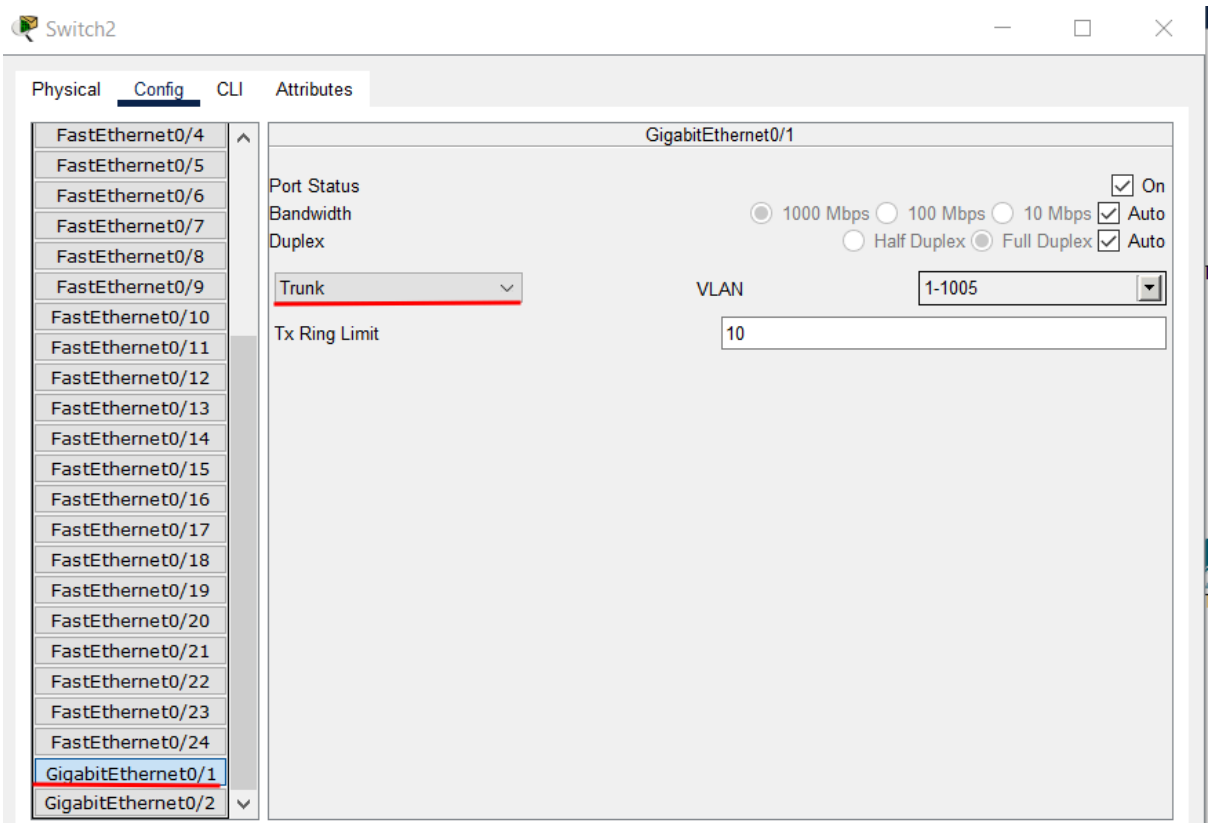
Now we have to configure our router, it needs to know some details how interfaces and where the connection is.



In the router settings, turn on the Gigabietherneto/o to our communication works.



We have to create the vlans in our router also.



In the switch that is connected to the router, we have to put in trunk in the output for the router.

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:42FF:FE8C:D121
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.3
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.20.5

Pinging 192.168.20.5 with 32 bytes of data:

Reply from 192.168.20.5: bytes=32 time<1ms TTL=127
Reply from 192.168.20.5: bytes=32 time<1ms TTL=127
Reply from 192.168.20.5: bytes=32 time<1ms TTL=127
Reply from 192.168.20.5: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Here I did a command to see my own IP, so then I tried to ping another pc in another vlan, and as we can see, we had success 😀.

The router allows communication between different VLANs because it knows the VLANs and their IP addresses.

Subinterfaces allow a single physical interface to function as multiple logical interfaces, and are often used for interVLANS routing.

VLANs use different IP address ranges so that each VLAN works as an independent network, enabling segmentation, proper routing, and efficient traffic management.